



ASTRON

RE-ENTRY FOR

Reusable Launch Vehicles

"Applied" Hypersonics

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HyperStructures 2026 · Talk 3 · Wed 2 Sep 2026, 15:05 · University of Nottingham

I · WHO IS SPEAKING

Astron Systems in one slide



Fully reusable

small launch vehicle — ~360kg to LEO, targeting marginal cost <\$400/kg long-term. Launch and return markets serviced.

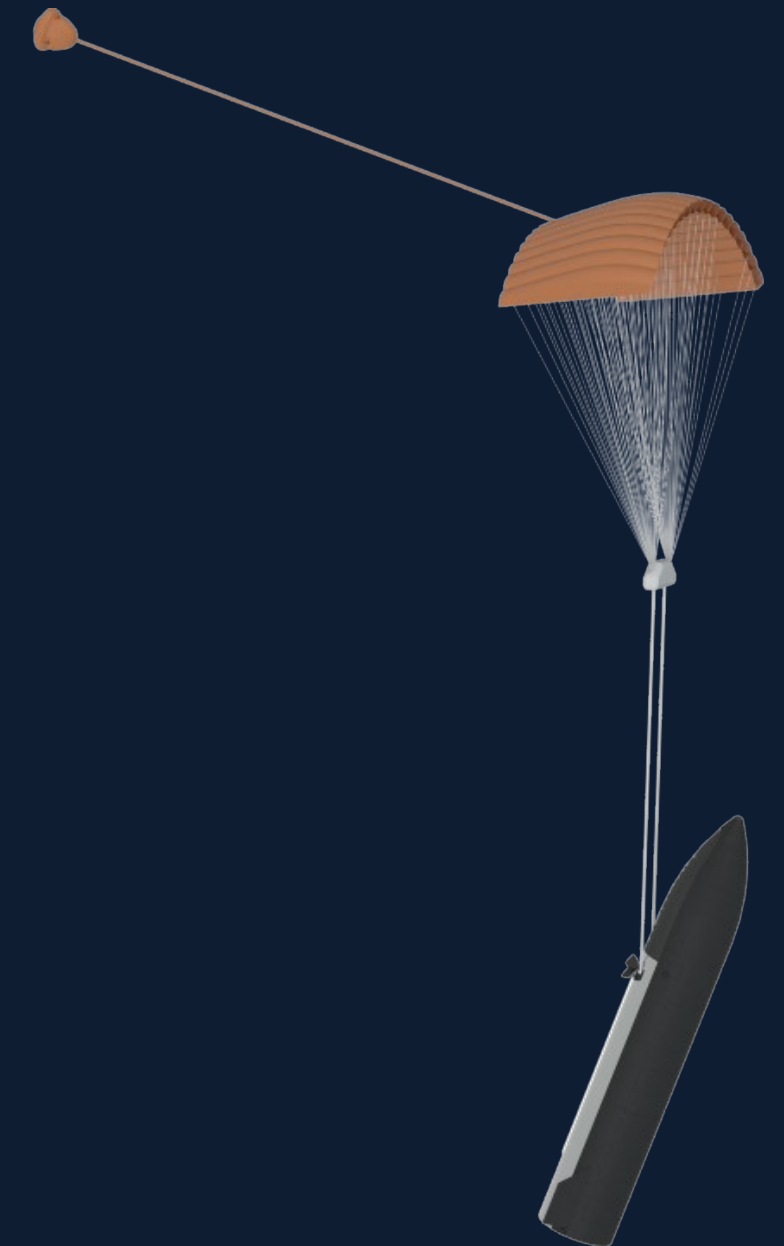
R&D in long-life time propulsion systems

electric-turbopump (HETP) propulsion; various ESA projects in propulsion disciplines

Low marginal cost

targeting high cadence and reliability, UK-sovereign

Venture-backed · dual focus: propulsion + aerothermodynamics · building the first fully reusable small launcher · HQ Harwell Campus · test site Westcott · Team Hypersonics UK



Why I am at a hypersonics-modelling workshop: re-entry modelling critical to understand the thermal and mechanical loads on an upper stage that we want to return, and then re-use. Descent by definition will cross the continuum-kinetic boundary.

I · THE STAKES

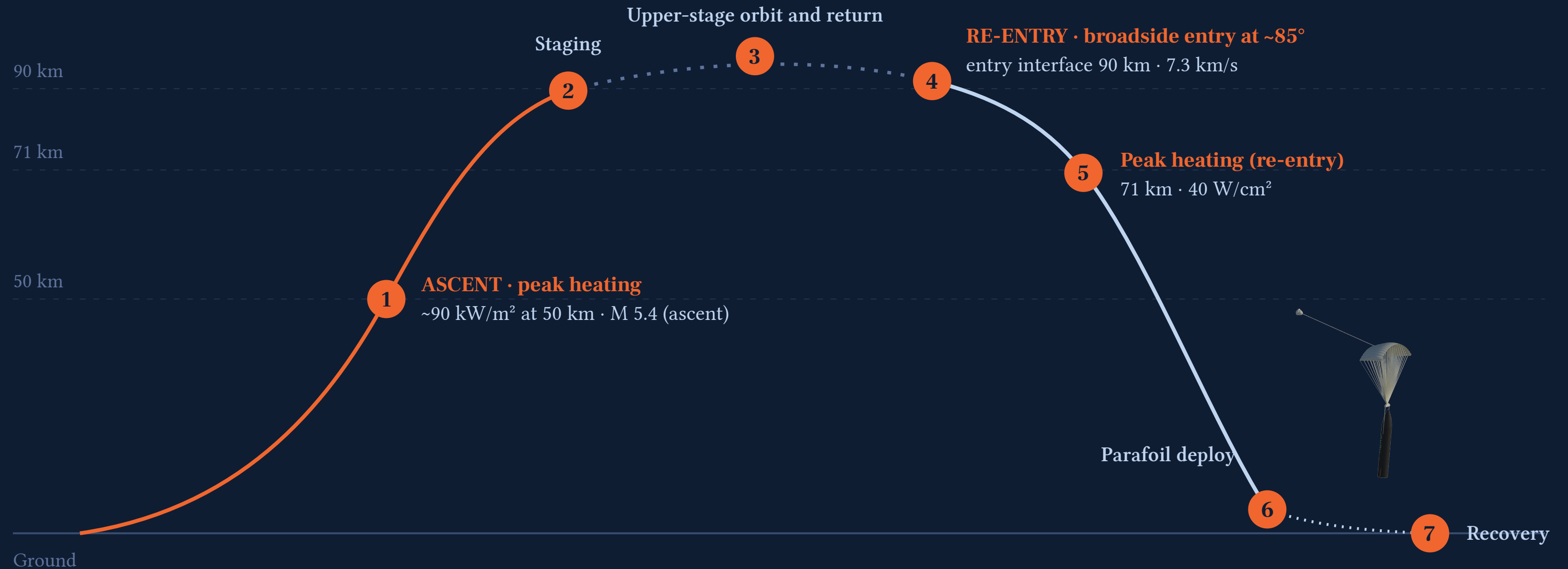
“Applied” Hypersonics – e.g. we have to fly it (eventually...)



Torch igniter firing on liquid methane / LOX · the hybrid-electric turbopump pumping cryogenics · a customer hot-fire hosted at our Westcott site.

I · THE FLIGHT WE HAVE TO CLOSE

Mission profile



Re-entry critical for economic feasibility



1

Reuse economics

The business case for a reusable stage only makes sense if it returns in a re-flyable condition — every lost stage has a strong impact on the marginal cost.

2

Aerothermal load sets TPS mass

Peak heating and integrated heat load size the thermal protection system. Mass set aside for a (heavy) TPS system is directly equivalent to lost payload mass.

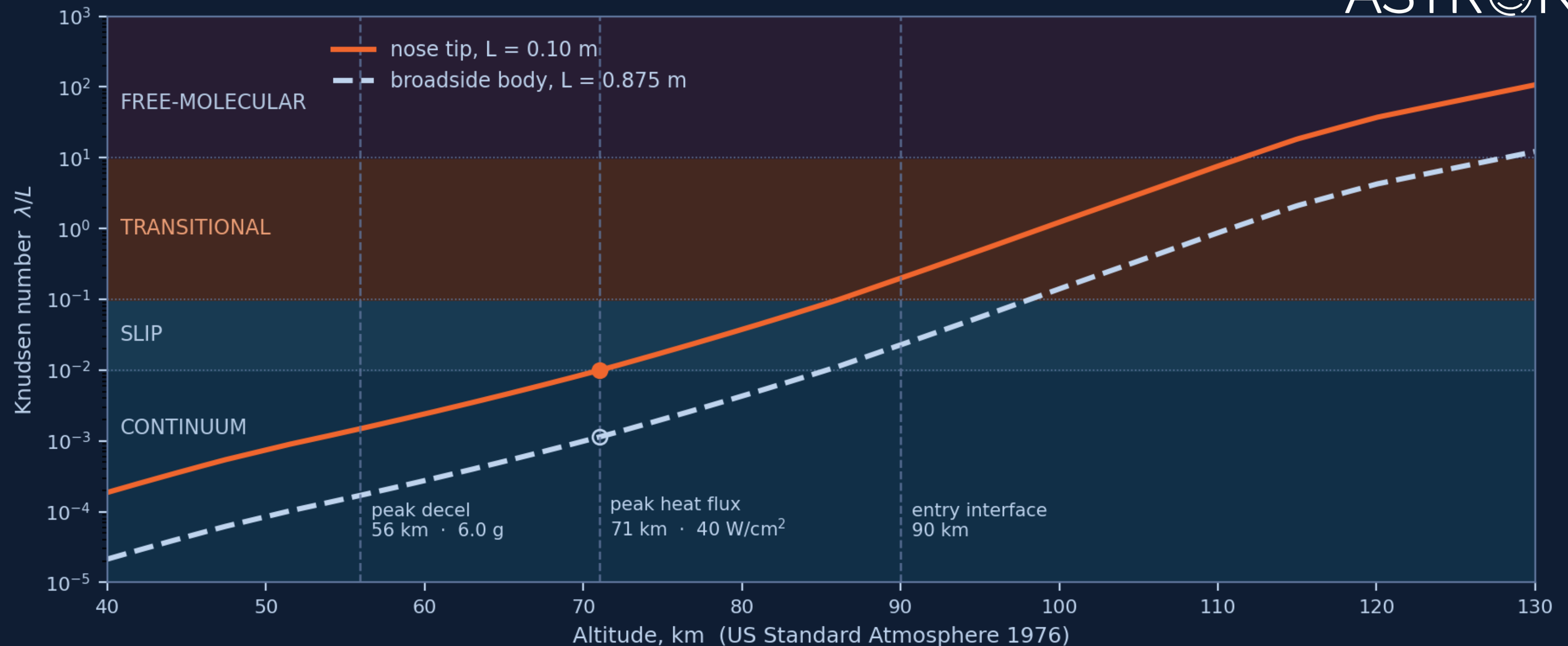
3

TPS mass sets payload

Over-conservative re-entry prediction → heavier TPS → less payload → worse unit economics. Under-prediction → loss of vehicle.

The prediction accuracy of the re-entry environment is therefore a direct lever on payload and cost.

Re-entry regimes for a reusable launch vehicle



The standard assumption is that peak heating for re-entry sits in the **transitional regime**. Computing it for our stage: the interface is crossed in transitional flow, but peak heating arrives in continuum — crossing the band costs only ~1% of the stage's velocity. The non-continuum physics that matters is embedded: shock structure, tip-scale slip, the separated lee-side.

The aerothermal environment to predict



Peak heat flux

Sets the TPS material limit and surface temperature. **40 W/cm² at 71 km** on the upper-stage end-of-life trajectory — 4.5× the ascent peak.

Integrated heat load

Sets TPS thickness / mass through the descent. **10.2 kJ/cm² integrated** — this is 21× the ascent load of 0.49 kJ/cm².

Not in a very (ideal) regime

Perfect-gas stagnation at the peak-heating point: ~19,000 K. Real-gas: nearer 6,500 K.

Aerodynamic + control coupling

Stability and trim of a recovering stage change with regime; drives GNC margins.

The value of better predictions



1

A 1 mm stainless airframe

The desired launch vehicle's skin is ~1 mm work-hardened 304 stainless (retains strength at temperature, tough at cryo, low cost — a thin, hot-capable airframe). Hardness is lost to annealing near 500 °C — far below any melt or ablation limit.

2

Ascent already a constraint for primary structures

At the ~90 kW/m² ascent peak a 1 mm tip reaches ~600 °C — annealed. The fix is 1.8 mm of local thickening, ~0.5 kg. The body never exceeds 82 °C.

3

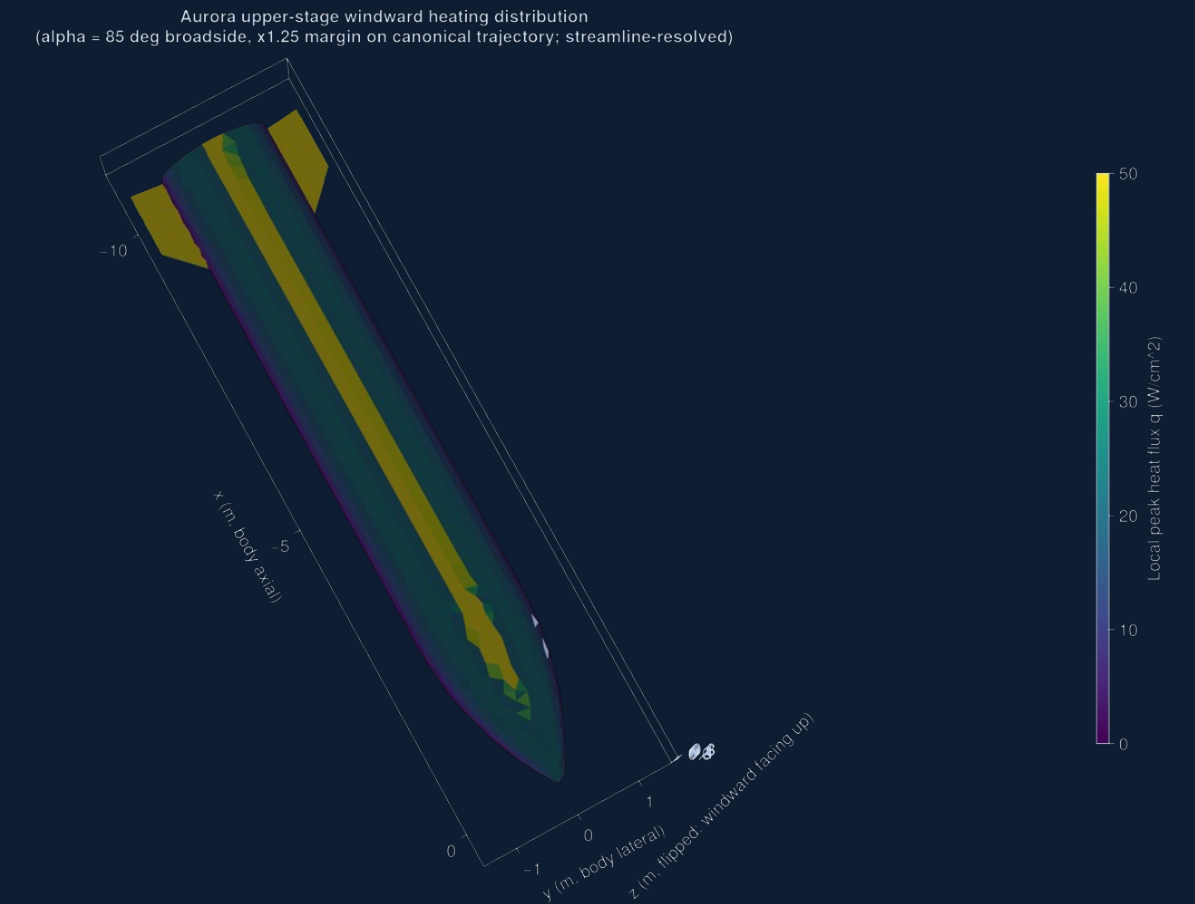
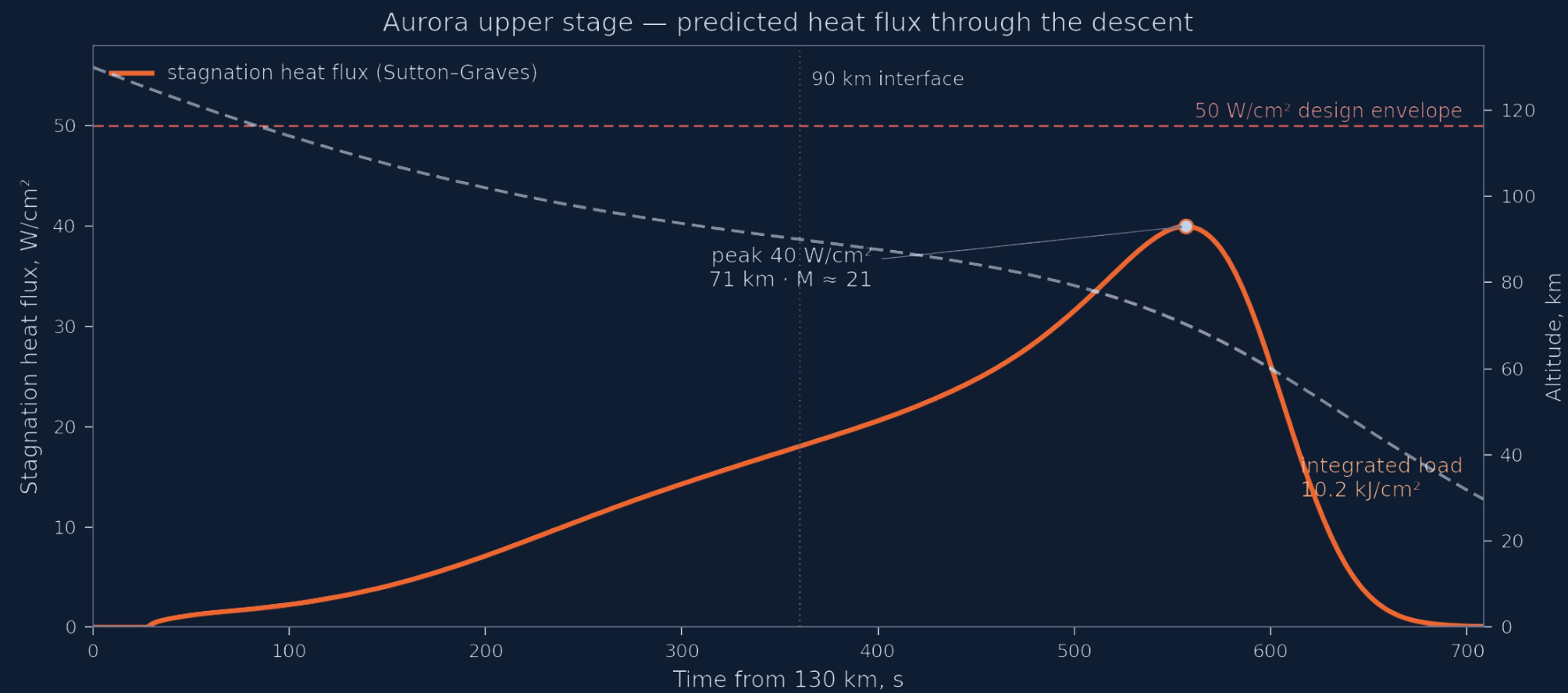
Prediction error has impact

Peak flux scales as $1/\sqrt{R_n}$: halve the tip radius and 90 becomes 127 kW/m² — local thickening stops working and a TPS begins. Re-entry is 4.5× the peak, 21× the load.

Prediction uncertainty can make otherwise ideal materials feasible or infeasible.

III · THE ENVIRONMENT, AS MODELLED

Current modelling outputs



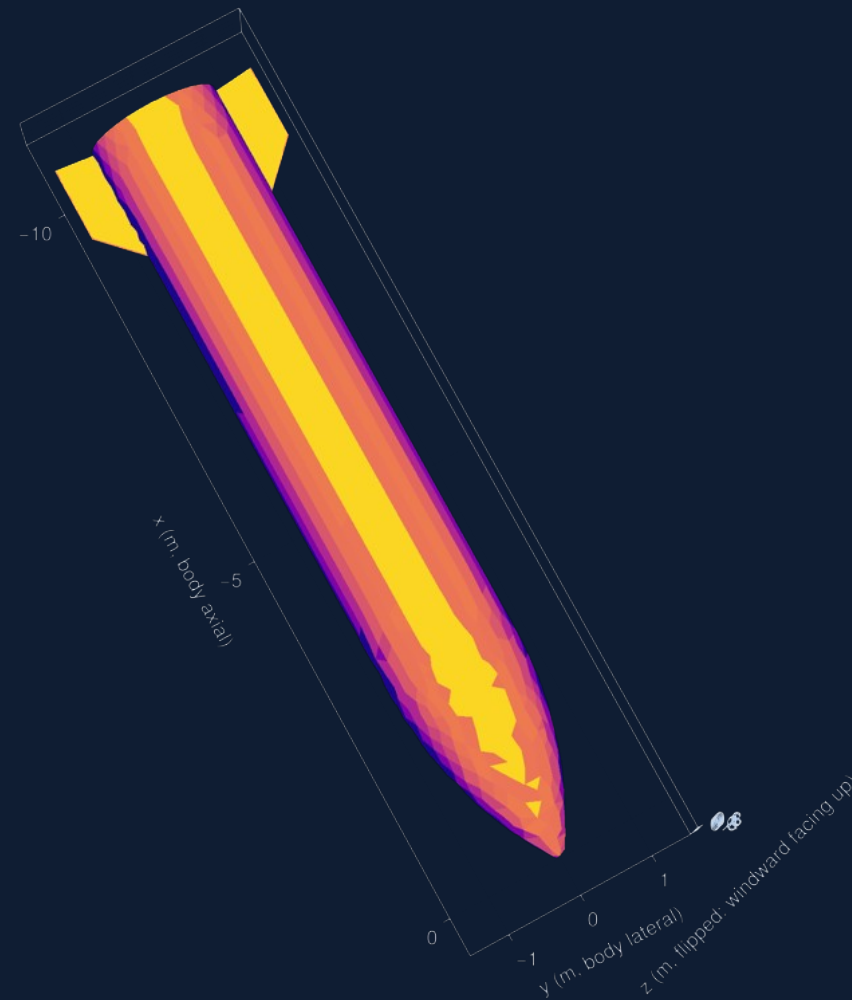
Left: stagnation heat-flux history through the descent — 40 W/cm² peak at 71 km against a 50 W/cm² design envelope; 10.2 kJ/cm² integrated. Right: windward heating distribution at 85° broadside — stagnation line along the spine; fin tips see stagnation-class flux.

III · FROM ENVIRONMENT TO HARDWARE

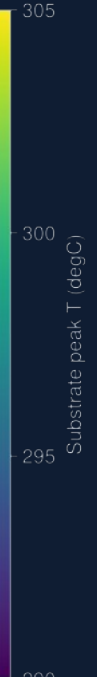
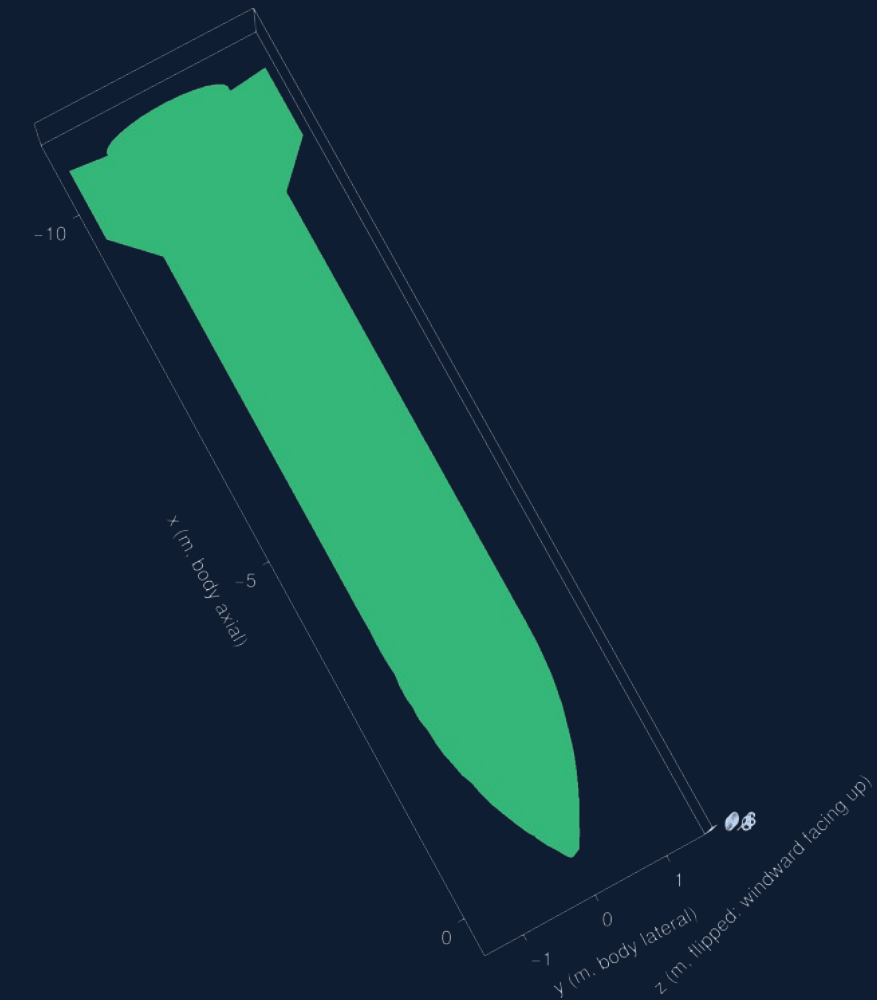
TPS sizing from the modelled environment



Optimal-design Saffil blanket thickness across Aurora upper-stage windward face
(per-band sized to substructure 300 degC under streamline-resolved heating)



Substrate peak temperature under OPTIMAL design (per-band Saffil sizing)
All windward bands sit at 300 degC by construction (design exhausts the substrate-T budget uniformly)



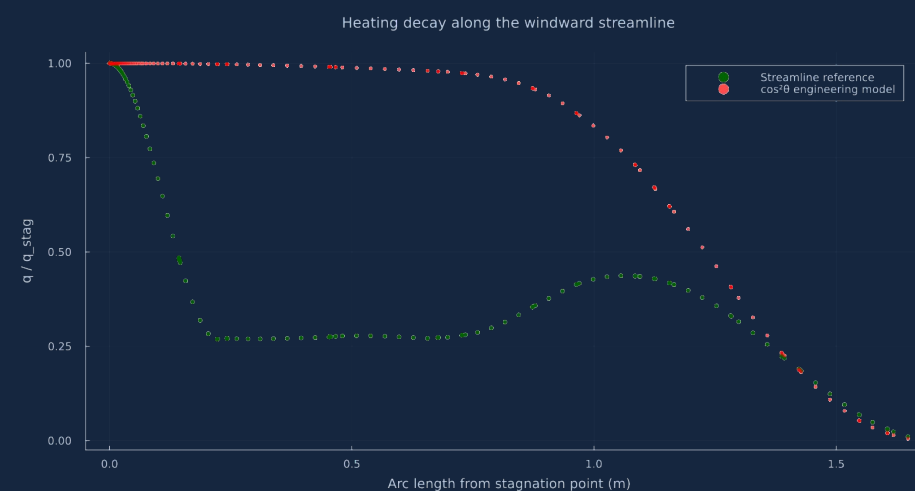
Per-zone blanket thickness across the windward face — 4.5 to 26 mm by local flux — and substructure peak temperature under the sized design: **all bands at the 300 °C limit by construction**. The environmental model can be coupled to the engineering sizing of the TPS

Capabilities and future direction



TODAY

- 1 · **Engineering correlations** — Fay–Riddell / Sutton–Graves family, full-trajectory sweeps
- 2 · **Distributed surface heating** — streamline tracing, reference-temperature methods; the CBAERO / SCARAB class of tool.
- 3 · **CFD / kinetic / DSMC** — RANS experience exists (Fluent / Star-CCM+). TPS sizing becomes a function of the best estimate of heating

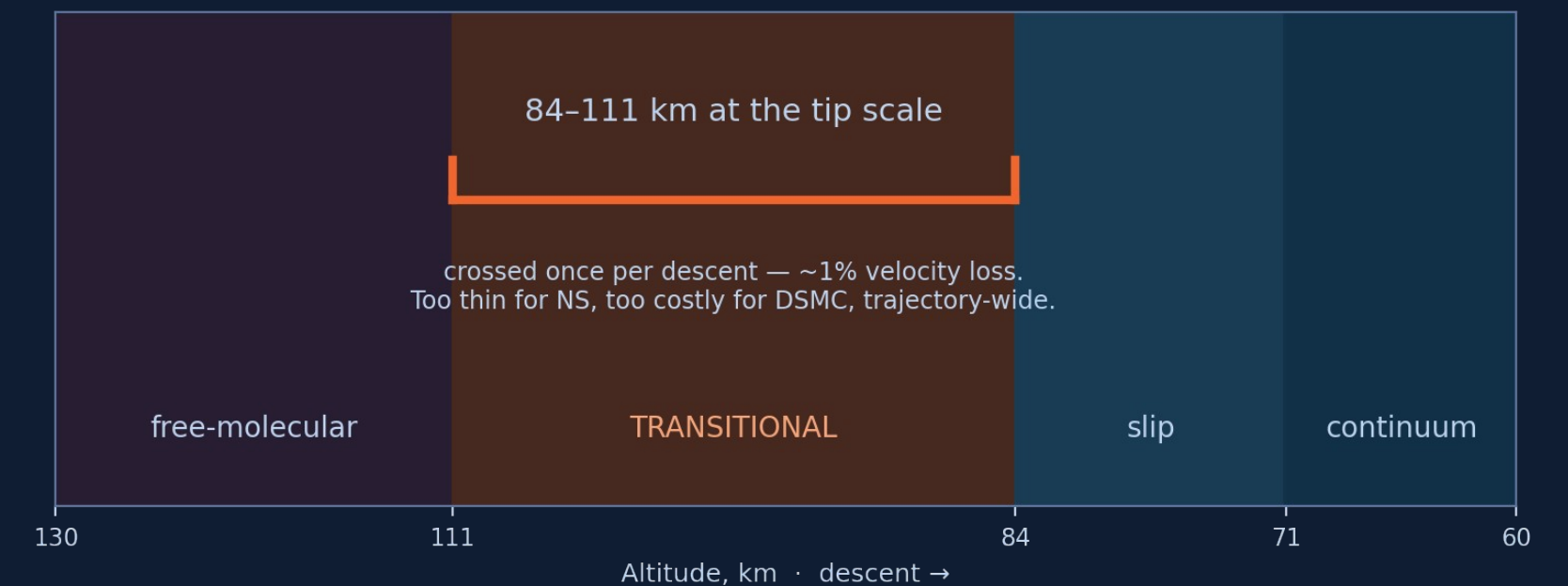


DIRECTION

Coupled aerothermal–TPS for re-flight life — understand mechanisms of wear.

Uncertainty on peak flux and integrated load, impact from tolerances — e.g., perturbations to re-entry trajectory, material tolerances, etc.

Hybrid transitional methods for the 84–111 km traverse.



Case studies of interest



1 · Entry interface, broadside

1.75 m cylinder at $\sim 85^\circ$ AoA, 7.3 km/s at 90 km — Kn 0.02–0.2 by length scale. Rarefied crossflow, separated lee wake; aero moments and trim authority — the attitude carried into the pulse.

2 · The peak-heating point

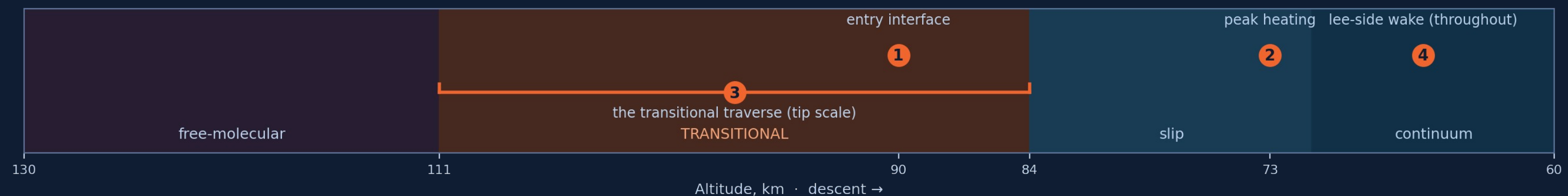
$M \approx 21$ at 73 km: continuum body, slip at the 0.1 m tip, bow shock a few λ thick regardless. Slip-BC validity and shock structure where uncertainty on heat flux *could* have the most impact.

3 · Crossing the transitional region

84–111 km at the tip scale — computationally too expensive to analyse trajectory-wide with current tools.

4 · Leeward and base flow

At broadside the separated lee-side is where continuum methods are least accurate — and for our launcher where the recovery system lives. Wake rarefaction, adjacent to decelerator work.





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Thank you

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Where better models pay off



- 1 Affordable transitional-regime aerothermal prediction across a full descent trajectory.
- 2 Quantified uncertainty on peak flux and integrated load — margin is mass.
- 3 Coupled aerothermal ↔ TPS response for repeated re-flight (life, wear mechanisms).
- 4 Regime-dependent aerodynamics for a recovering stage — trim, stability, control authority.
- 5 Fast surrogates good enough to sweep a design space.